








Blockchain Applications

Lecture 2: Stylized Facts, Efficiency and Bubbles



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
Master Blockchain for Business (MBB), FABIZ
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Opening: Statistical Behaviour of Crypto Prices



Eugene F. Fama. Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

"A market in which prices always 'fully reflect' available information is called 'efficient.'"
— *Fama (1970)*

- Bitcoin's annualised volatility in 2018–2026 is 64.1%, against 19.2% for the S&P 500
- Crashes: Bitcoin had 17 days beyond 4 standard deviations in 3 183 days, against 0.2 expected under the Normal distribution
  - worst day, 12 March 2020: a log return of -46.5%
- Volatility clusters: the Ljung–Box statistic $Q(10)$ is 212.7 for squared returns, against 20.2 for returns
- Robust variance-ratio tests do not reject a random walk at 2 to 10 days in any sub-period
  - not rejecting a random walk is not proof of efficiency
- The log-periodic power law (LPPL) fits the 2017 bubble with $R^2 \approx 0.98$, yet the estimated crash date moves with the window

Central chart: Efficiency Through Time

Lecture Question and Outline

- **Question**: how do crypto prices behave, and what does that mean for a business that holds, accepts or prices crypto?
  - a treasurer, an exchange or a fund needs numbers, not narratives
- **Outline**
  - prices and returns on a 7-day market: Lecture 1 described levels and trends; today the same prices become returns
  - the shape of returns: heavy tails, rare but frequent crashes
  - volatility through time, and the link with equities and gold
  - market efficiency: can past prices predict returns?
  - bubbles, drawdowns and network value (Metcalfe's law)
- Lecture 4: models for volatility and tail risk: generalised autoregressive conditional heteroskedasticity (GARCH), value at risk (VaR), expected shortfall (ES)

Learning Outcomes

- Compute log returns and annualise volatility with the right number of days
- Read a summary table: volatility, skewness, kurtosis, worst day
- Use a histogram and a quantile–quantile (QQ) plot to show why the Normal distribution understates crash risk
- Interpret rolling volatility and rolling correlation charts
- Explain weak-form efficiency and read a variance-ratio test
- Measure drawdowns and judge bubble and network-value claims with data

Abbreviations

ACF autocorrelation function
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CI confidence interval
COVID-19 coronavirus disease 2019
CRIX CRypto IndeX (Trimborn & Härdle (2018))
EODHD EOD (end-of-day) Historical Data, data vendor
ES expected shortfall
EWMA exponentially weighted moving average
FABIZ Faculty of Business Administration in Foreign Languages, ASE
FTX crypto exchange (company name), bankrupt in November 2022
GARCH generalised autoregressive conditional heteroskedasticity
i.i.d. independent and identically distributed
IDA Institute Digital Assets
JB Jarque–Bera (test statistic)
LPPL log-periodic power law
LPPLS log-periodic power law singularity
MBB Master Blockchain for Business
QQ quantile–quantile
S&P 500 Standard & Poor's 500 US stock index
sd standard deviation
SE standard error
SFM Statistics of Financial Markets (master course; Quantlet prefix)
USD US dollar
UTC Coordinated Universal Time
VaR value at risk
VR variance ratio

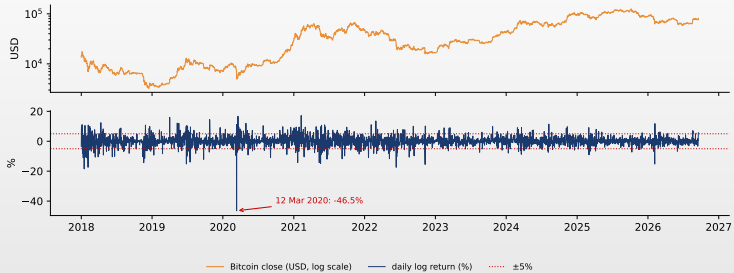
Reading and Tools

- Core reading: Trimborn & Härdle (2018), *CRIX: an index for cryptocurrencies*; Peterson (2018), *Metcalfe's law as a model for Bitcoin's value*
- Video units on Quantinar (course Blockchain Analytics)
  - Blockchain Mechanism and Distributional Characteristics of Cryptos
  - Crypto Currency Returns
- Data
  - daily closes of Bitcoin, Ethereum, the S&P 500 and gold, 2018 – 18 September 2026 (EODHD)
  - on-chain transactions of Bitcoin, 2014–2025 (ETH Chronicles project, Lecture 1)
- Further: Pele et al. (2023), *Are cryptos becoming alternative assets?*

Log Returns and Annualisation

- Daily **log return** in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$; P_t: the close on day t
- Annual volatility from daily data: $\sigma_a = s\sqrt{A}$
  - s: the standard deviation of daily returns; A: the number of returns per year
  - crypto trades every day: $A = 365$; stocks: $A \approx 252$ trading days
- **Worked example**: Bitcoin 2018–2026, $s = 3.355\%$ a day
  - right: $\sqrt{365} \times 3.355 = 19.10 \times 3.355 = 64.1\%$ a year
  - wrong: $\sqrt{252} \times 3.355 = 53.3\%$, an understatement of 17%
- Rule: annualise each asset with its own calendar; compare assets only after that

Bitcoin Prices and Log Returns, 2018–2026



- Top: daily close in USD, log scale; bottom: daily log returns in %, 1 January 2018 – 18 September 2026, $n = 3\,183$
- The price trends and swings; the returns stay around zero, with bursts: 340 days (10.7%) beyond $\pm 5\%$
- Worst day 12 March 2020: -46.5%; best day 8 February 2021: $+17.2\%$
- Log returns add up: the sum of the 3 183 daily returns, 174.3, equals $100 \ln(P_{\text{end}}/P_{\text{start}})$

BA_L2_returns_path

Moments of the Return Distribution

- For n daily returns $r_1, \dots, r_n$:

$$\bar{r} = \frac{1}{n}\sum_{t=1}^{n} r_t, \qquad s^2 = \frac{1}{n-1}\sum_{t=1}^{n}(r_t - \bar{r})^2$$

$$\hat{S} = \frac{1}{n}\sum_{t=1}^{n}\left(\frac{r_t - \bar{r}}{s}\right)^3, \qquad \hat{K} = \frac{1}{n}\sum_{t=1}^{n}\left(\frac{r_t - \bar{r}}{s}\right)^4 - 3$$

  - $\bar{r}$: mean; s: standard deviation (volatility); $\hat{S}$: skewness; $\hat{K}$: excess kurtosis
  - Normal distribution: $S = 0$ and $K = 0$

- **Jarque–Bera (JB) test** of normality:

$$\mathrm{JB} = \frac{n}{6}\left(\hat{S}^2 + \frac{\hat{K}^2}{4}\right) \sim \chi^2_2 \text{ under } H_0\text{: Normal}$$

  - reject normality at 5% if $\mathrm{JB} > 5.99$
  - 2018–2026: Bitcoin 29 620, Ethereum 16 593, S&P 500 20 462, gold 10 041: normality is rejected for all four

Worked Example: Jarque–Bera for Bitcoin

- Bitcoin, 2018–2026: $n = 3\,183$, $\hat{S} = -0.945$, $\hat{K} = 14.825$

$$\mathrm{JB} = \frac{3\,183}{6}\left(0.945^2 + \frac{14.825^2}{4}\right) = 530.5 \times (0.893 + 54.945) \approx 29\,620$$

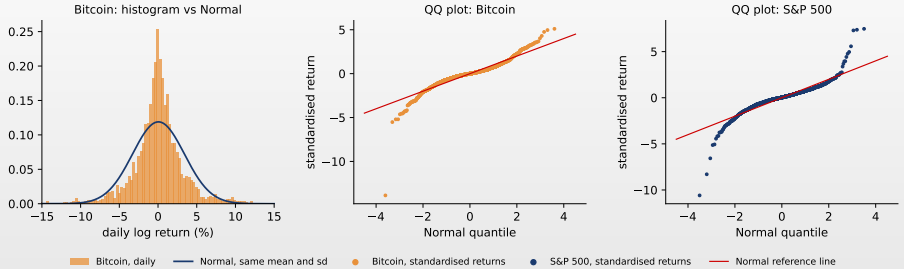
- 5% critical value of χ^2_2: 5.99; p-value $< 10^{-300}$
- Almost all of the statistic comes from kurtosis (54.9 of 55.8): the problem is the tails, not the asymmetry
- Caution: with n in the thousands, JB rejects even mild deviations; the size of $\hat{K}$ matters more than the p-value

Four Assets Side by Side, 2018–2026

	n	sd (%)	ann. vol. (%)	skewness	excess kurtosis	worst day (log %)	days $\lvert r\rvert > 5\%$ (%)
Bitcoin	3 183	3.36	64.1	−0.95	14.8	−46.5	10.7
Ethereum	3 183	4.41	84.3	−0.83	11.1	−55.1	17.8
S&P 500	2 189	1.21	19.2	−0.65	14.9	−12.8	0.6
Gold	2 268	1.07	16.9	−0.55	10.2	−11.1	0.3

- Daily log returns, 1 January 2018 – 18 September 2026, each asset on its own calendar
  - crypto: every day, $A = 365$; S&P 500 and gold: Monday–Friday closes, $A = 252$
  - **skewness**: 0 for a symmetric distribution; negative: the left tail (losses) is longer
  - **excess kurtosis**: 0 for the Normal distribution; positive: more extreme days than the Normal predicts
- Interpretation
  - Bitcoin is 3.3 times as volatile as the S&P 500; it moves by more than 5% on one day in nine
  - the worst day of Bitcoin and Ethereum was the same: 12 March 2020 (COVID-19 crash)
  - all four are skewed to the left and heavy-tailed

Fit of the Normal Distribution



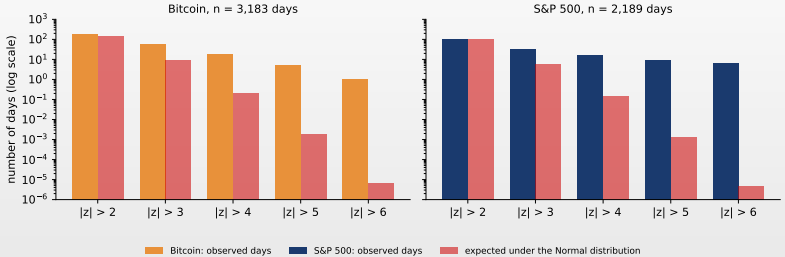
- Left: too many quiet days in the centre and too many extreme days in the tails, too few in between
- **QQ plot**: sorted returns against the quantiles a Normal distribution would give; Normal data lie on the red line
- Both QQ plots bend away at the ends: heavy tails in crypto *and* in equities

BA_L2_hist_qq ▶ Interactive

Frequency of Crashes

- Standardise: $z_t = (r_t - \bar{r})/s$, the number of standard deviations of each day
- Under the Normal distribution
  - $P(|z| > 3) = 0.27\%$; $P(|z| > 4) = 0.0063\%$
  - in 3 183 days we would expect 0.2 days beyond 4 standard deviations
- Bitcoin, 2018–2026: **17 days** beyond 4 standard deviations
- Worst day, 12 March 2020: log return -46.5%, i.e. $z = -13.9$
  - as a simple return: $e^{-0.465} - 1 = -37.2\%$ of the value held
  - under the Normal: $P(z \leq -13.9) \approx 5 \times 10^{-44}$, i.e. once in 2×10^{43} days
- Business lesson: a risk model built on the Normal distribution will be surprised many times; Lecture 4 fixes this

Crash Days: Observed against the Normal Distribution



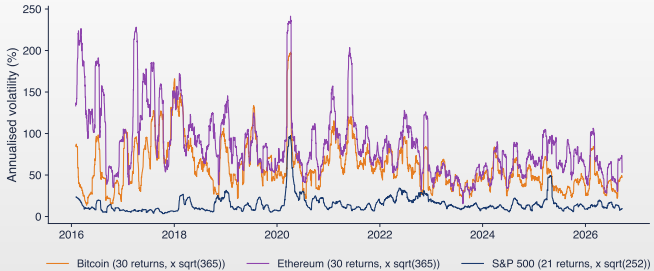
- Days with $|z_t| > k$, $k = 2, \ldots, 6$, in 2018–2026, against $n \times P(|z| > k)$ under the Normal distribution (log scale)
- $k = 2$: Bitcoin 185 days against 145 expected, S&P 500 99 against 100: the Normal fits the moderate days
- $k = 4$: Bitcoin 17 against 0.20 (84 times as many), S&P 500 15 against 0.14 (108 times)
- $k = 6$: Bitcoin 1 day, S&P 500 6 days; the Normal expects one such day in 507 million
- The ratio of observed to expected grows with k: the tails, not the centre, break the Normal distribution

BA_L2_crash_counts

Worked Example: Expected Crash Days under the Normal

- Bitcoin, 2018–2026: $n = 3\,183$ daily returns
- Step 1, $k = 3$: $P(|z| > 3) = 2 \times 0.00135 = 0.00270$
  - expected days $= 3\,183 \times 0.00270 = 8.6$; observed 59, seven times as many
- Step 2, $k = 4$: $P(|z| > 4) = 2 \times 3.17 \times 10^{-5} = 6.33 \times 10^{-5}$
  - expected days $= 3\,183 \times 6.33 \times 10^{-5} = 0.20$; observed 17, 84 times as many
- Step 3, waiting time between two days with $|z| > 4$
  - under the Normal: $1/(6.33 \times 10^{-5}) = 15\,787$ days, about 43 years
  - observed: $3\,183/17 = 187$ days, about six months
- Reading: a risk limit built on Normal tail probabilities treats a once-in-six-months event as a once-in-43-years event

Volatility Through Time



- Volatility on about one month: 30 daily returns $\times\sqrt{365}$ for crypto, 21 returns $\times\sqrt{252}$ for the S&P 500
- Medians: 54% (Bitcoin), 77% (Ethereum), 12% (S&P 500); all three peak in March–April 2020
- Bitcoin's volatility trends down, but it was below that of the S&P 500 on only 0.1% of the days

SFM_ch14_asset_class ▶ Interactive

Autocorrelation and the Ljung–Box Test

- **Sample autocorrelation** at lag k:

$$\hat{\rho}(k) = \frac{\sum_{t=k+1}^{n}(r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^{n}(r_t - \bar{r})^2}$$

  - under independence, $\hat{\rho}(k)$ lies within $\pm 1.96/\sqrt{n}$ with probability 95%
  - autocorrelation function (ACF) of r_t: is the direction predictable? ACF of r_t^2: is the size of the move predictable?

- **Ljung–Box** test of the first m autocorrelations together:

$$Q(m) = n(n+2)\sum_{k=1}^{m}\frac{\hat{\rho}(k)^2}{n-k} \sim \chi^2_m \text{ under } H_0\text{: no autocorrelation}$$

  - Bitcoin daily log returns, 18 September 2014 – 18 September 2026, $n = 4\,384$, $m = 10$ lags
  - $Q = 20.2$ (p-value 0.028) for r_t; $Q = 212.7$ for r_t^2; the 5% critical value of χ^2_{10} is 18.3
  - weak dependence in returns, strong dependence in squared returns

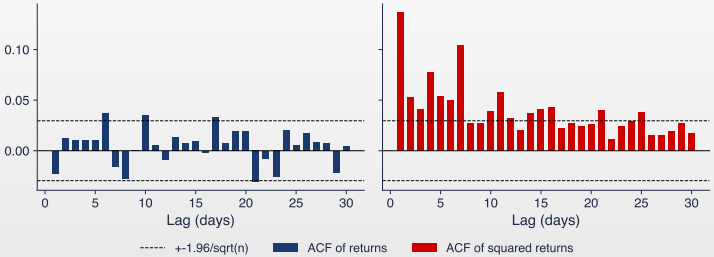
Worked Example: Ljung–Box with Two Lags

- Bitcoin daily log returns since 2014, $n = 4\,384$; $Q(2) = n(n+2)\left[\hat{\rho}(1)^2/(n-1) + \hat{\rho}(2)^2/(n-2)\right]$
- Returns: $\hat{\rho}(1) = -0.0223$, $\hat{\rho}(2) = 0.0121$

$$Q(2) = 4\,384 \times 4\,386 \times \left(\frac{0.000498}{4\,383} + \frac{0.000147}{4\,382}\right) = 2.83$$

  - below 5.99, the 5% critical value of χ^2_2 (p-value 0.24): we do not reject "no autocorrelation"
- Squared returns: $\hat{\rho}(1) = 0.1367$, $\hat{\rho}(2) = 0.0527$, so $Q(2) = 94.2$ (p-value 4×10^{-21}): rejected
  - the first term alone, $\approx n\,\hat{\rho}(1)^2 = 4\,384 \times 0.0187 = 81.9$, decides the test
- One lag at a time: the 95% band is $\pm 1.96/\sqrt{4\,384} = \pm 0.030$; $\hat{\rho}(1)$ of returns lies inside it, that of squared returns far outside

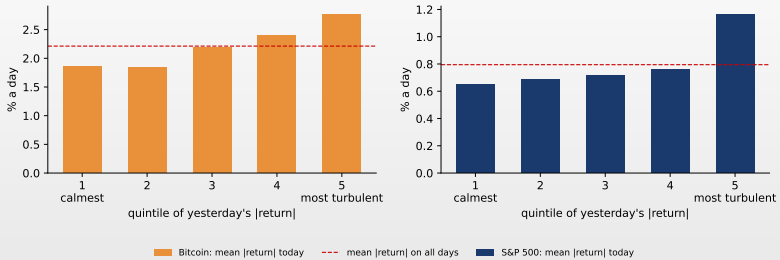
Volatility Clusters



- **Autocorrelation** at lag k: the correlation between today's value and the value k days ago; dashed: the 95% band of pure noise
- Returns (left): almost no autocorrelation; the *direction* of tomorrow's move is hard to predict
- Squared returns (right): clearly positive; the *size* of tomorrow's move is predictable: calm days follow calm days, turbulent days follow turbulent days
- This is **volatility clustering**, the basis of the GARCH models of Lecture 4

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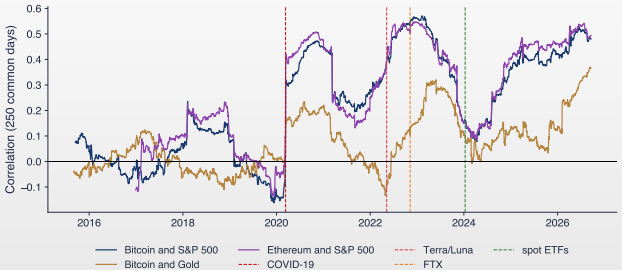
Absolute Returns after Calm and Turbulent Days



- Days of 2018–2026 sorted into five equal groups (quintiles) by yesterday's absolute return; bars: today's mean absolute return
- Bitcoin: 1.86% after the calmest days, 2.77% after the most turbulent, 1.49 times as much; all days 2.21%
- S&P 500: 0.65% and 1.17%, 1.80 times as much; all days 0.79%
- Yesterday's size says something about today's size, not about its sign: the volatility clustering seen in the ACF of r_t^2

BA_L2_calm ▶ Interactive

Correlation of Bitcoin with Equities



- Correlation of daily returns over the last 250 common trading days, about one year (method: Appendix)
- Before 2020: no link with the S&P 500 (0.02); after: 0.39 on average, peak 0.57 in February 2023
- Since 2020 Bitcoin co-moves with equities, as a **risk asset** would; in 2022 both fell while interest rates rose

SFM_ch14_correlation ▶ Interactive

Weak-Form Efficiency and the Variance Ratio

- Fama (1970): in a **weak-form efficient** market, past prices do not help to predict future returns
  - if they did, traders would exploit the pattern until it disappeared
- **Variance ratio** (VR) Lo & MacKinlay (1988): compare the variance of q-day returns with q times the daily variance

$$\mathrm{VR}(q) = \frac{\mathrm{Var}(r_t + r_{t-1} + \cdots + r_{t-q+1})}{q\,\mathrm{Var}(r_t)}$$

  - random walk (no predictability): $\mathrm{VR} = 1$
  - $\mathrm{VR} > 1$: trends (momentum); $\mathrm{VR} < 1$: reversals (mean reversion)
  - a test statistic Z^*, robust to volatility clustering, rejects $\mathrm{VR}(q) = 1$ at 5% if $|Z^*| > 1.96$
- The VR test checks one implication of weak-form efficiency: no linear autocorrelation of returns up to horizon q

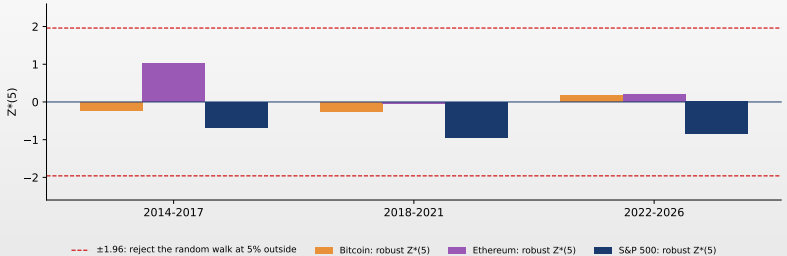
Variance-Ratio Tests

	VR(2)	VR(5)	VR(10)	$Z^*(5)$ (p-value)
Bitcoin	0.977	0.991	1.027	−0.19 (0.853)
Ethereum	0.988	1.067	1.137	1.17 (0.241)
S&P 500	0.875	0.849	0.813	−1.28 (0.201)

- Daily log returns since 2014 (Ethereum since 2016) to 18 September 2026
- Interpretation
  - all variance ratios are close to 1; no test rejects $\mathrm{VR}(5) = 1$ at 5%
  - the same holds in each sub-period 2014–2017, 2018–2021, 2022–2026
  - we do not detect the tested deviation from a random walk at these horizons; this does not prove efficiency
  - profitability after trading costs and risk adjustment is a separate question

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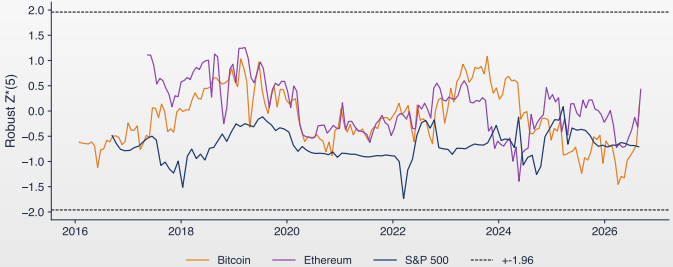
Variance Ratios by Sub-Period



- Robust $Z^*(5)$ in three sub-periods, to 18 September 2026 (Ethereum from 2016); dashed: ± 1.96
- Bitcoin -0.23, -0.25, $+0.18$; Ethereum $+1.03$, -0.04, $+0.20$; S&P 500 -0.68, -0.95, -0.85
- No bar crosses ± 1.96: we do not reject the random walk in any sub-period
- S&P 500: $\mathrm{VR}(5)$ is 0.92, 0.78 and 0.92, below 1 each time; mild mean reversion of an index, not significant

BA_L2_vr_periods ▶ Interactive

Efficiency Through Time



- Robust $Z^*(5)$ on windows of 500 days, one window every 21 days; dashed: ± 1.96
- No 500-day window rejects $\mathrm{VR}(5) = 1$ for Bitcoin at 5%
- Early studies found Bitcoin inefficient before 2013–2016 Urquhart (2016); markets became more efficient as they grew Tran & Leirvik (2020)

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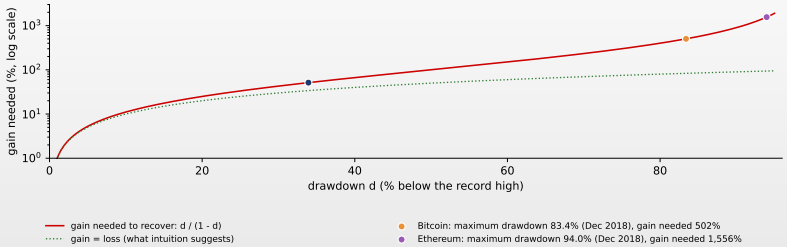
Interpreting the Efficiency Tests

- Not rejecting $\mathrm{VR} = 1$: no detectable linear predictability of returns from past returns at these horizons
  - other patterns (nonlinear, or conditional on volume or news) are not tested
  - even full weak-form efficiency says nothing about whether the price is *right*
- A price can be unpredictable and still far from any fundamental value
  - Bitcoin has no cash flows: no dividends, no profits to discount
  - so "fundamental value" must come from use: payments, settlement, collateral, network size
- Two questions remain
  - how far can the price fall from a peak? (drawdowns)
  - can we detect a bubble, and does network activity justify the price? (log-periodic power law (LPPL), Metcalfe)

Drawdown from the Running Maximum

- **Drawdown** on day t: $D_t = P_t / \max_{s \le t} P_s - 1$, the loss from the last record high
  - **maximum drawdown**: the deepest fall; **time under water**: the days until a new record
- A loss of d needs a gain of $d/(1-d)$ to recover
  - $d = 50\%$: $+100\%$; $d = 75\%$: $+300\%$; $d = 83.4\%$: $+502\%$
- Why managers care: a firm holding crypto on its balance sheet reports the drawdown, not the average return

The Gain Needed to Recover



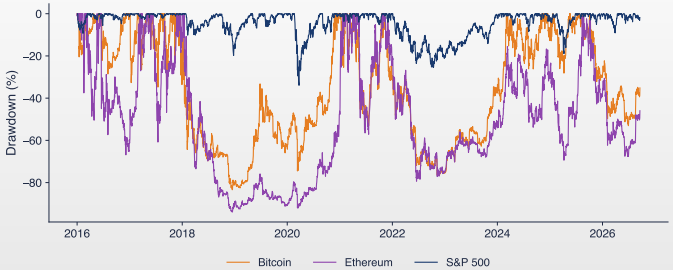
- Red: the gain $d/(1-d)$ that recovers a drawdown d, log scale; dotted green: a gain equal to the loss
- Maximum drawdowns of the daily closes since 2016
  - S&P 500: 33.9% (March 2020), recovered by a gain of 51%
  - Bitcoin: 83.4% (December 2018), gain needed 502%; Ethereum: 94.0% (December 2018), 1 556%
- Every extra point of loss needs more than one point of gain; beyond a loss of 50% the gain needed exceeds 100%

BA_L2_recovery

Worked Example: a Drawdown and Its Recovery

- Bitcoin: peak on 8 November 2021 at 67 567 USD; trough on 21 November 2022 at 15 787 USD
- Step 1, depth: $d = 1 - 15\,787/67\,567 = 76.6\%$
- Step 2, gain needed: $d/(1-d) = 0.766/0.234 = 328\%$, the same as $67\,567/15\,787 - 1$
- Step 3, time: new record on 4 March 2024, 847 days after the peak and 469 days after the trough
- Step 4, speed of the recovery: $4.28^{1/1.28} - 1 = 210\%$ a year from the trough
  - at a steady 50% a year the same recovery would take $\ln 4.28 / \ln 1.5 = 3.6$ years
- Reading: the time under water depends on how fast prices rise after the trough, which no one knows at the trough

Drawdowns of Bitcoin, Ethereum and the S&P 500



- Daily closes since 2016; maximum drawdowns: Bitcoin −83.4%, Ethereum −94.0%, S&P 500 −33.9%
- Bitcoin spent 76% of the days more than 10% below its record; the S&P 500 16%

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Bubbles and Crashes

- **Bubble**: a price far above any fundamental value, sustained by the expectation of selling higher
  - Cheah & Fry (2015): the price of Bitcoin shows speculative bubbles
- The idea of Johansen, Ledoit & Sornette (2000): imitation between traders creates **positive feedback**
  - prices grow *faster than exponentially*: the growth rate itself grows
  - the crash hazard rises as the price approaches a **critical time** t_c
  - the hierarchy of traders (individuals, funds, markets) adds oscillations that speed up near t_c
- The result is the **log-periodic power law** (LPPL), next slide



Didier Sornette, co-author of the LPPL model

Photo: Didier Sornette (2012); CC BY-SA 3.0 DE; Wikimedia Commons

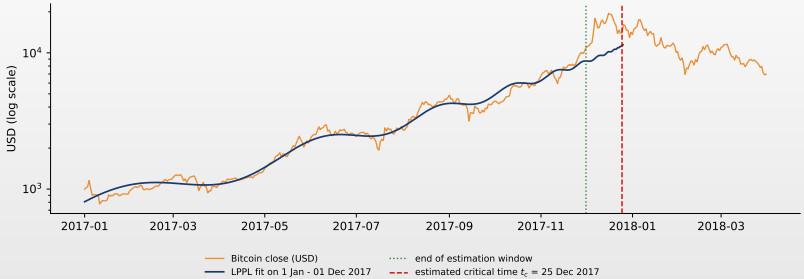
The LPPL Model

- Log price before the critical time, $t < t_c$; the model describes $\mathrm{E}[\ln P_t]$, and ε_t is the error:

$$\ln P_t = A + B\,(t_c - t)^m + C\,(t_c - t)^m \cos\big(\omega \ln(t_c - t) - \phi\big) + \varepsilon_t$$

- Parameters
  - t_c: critical time, the end of the bubble regime implied by the fitted model, not a guaranteed crash date; A: the log price reached at t_c
  - $B < 0$ and $0 < m < 1$: super-exponential growth; $(t_c - t)^m$ shrinks to 0 as $t \to t_c$
  - C: amplitude of the oscillations; ω: their log-frequency; ϕ: their phase
- **Log-periodic**: the peaks of the cosine come at times t_k with $(t_c - t_{k+1})/(t_c - t_k) = e^{-2\pi/\omega}$
  - the gaps between peaks shrink geometrically: the oscillations accelerate towards t_c
- Constraints in the estimation: $0.1 \le m \le 0.9$, $6 \le \omega \le 13$, t_c after the last observation
- Bubble filter on the fitted parameters: $\hat{B} < 0$ and $|\hat{C}|/|\hat{B}| < 1$

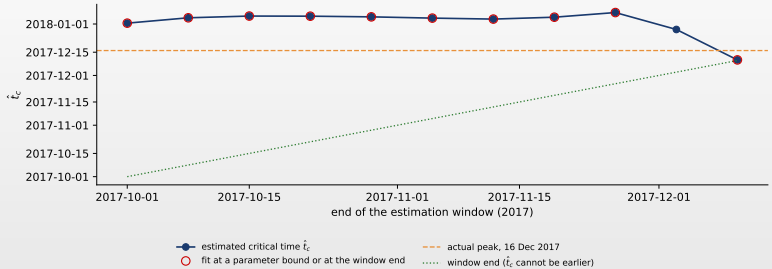
Bitcoin 2017: an LPPL Fit



- Daily log close, window 1 January – 1 December 2017 ($n = 335$); two-step estimation, least squares inside a search over (t_c, m, ω) (Appendix)
- Estimated t_c = 25 December 2017; the actual peak came on 16 December 2017 (19 497 USD), followed by a fall of 83%
- $R^2 = 0.98$ on a window chosen after the crash: a good retrospective fit, not a dated forecast (next slide)

BA_L2_lppl_btc2017 ▶ Interactive

$\hat{t}_c$ across Estimation Windows



- Eleven windows, all starting 1 January 2017, ending weekly from 1 October to 10 December 2017
- Nine of eleven fits sit at the bounds $m = 0.9$, $\omega = 13$ (red circles): the parameters are poorly identified; their $\hat{t}_c$ lie between 1 and 7 January 2018
- Only the window ending 3 December gives an interior fit ($\hat{t}_c$ = 28 December 2017)
- Window ending 10 December: $\hat{t}_c$ equals the last observation, a degenerate solution

BA_L2_lppl_tc ▶ Interactive

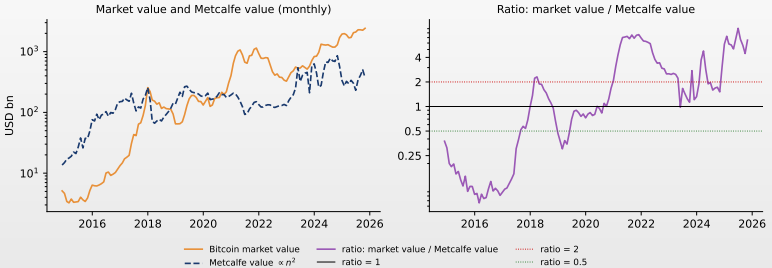
Network Value: Metcalfe's Law

- **Metcalfe's law**: the value of a network grows with the square of its users n
  - n users can form about $n^2/2$ connections; each connection adds value
  - used to value telecom and social networks; applied to Bitcoin by Peterson (2018)
- As a regression on logs:

$$\ln M_t = a + b \ln n_t + u_t, \qquad \text{Metcalfe: } b = 2$$

  - M_t: market value; n_t: a measure of use, here Bitcoin transactions per day; u_t: the error
  - b is an **elasticity**: 1% more use goes with b% more value
- Data: monthly means, November 2014 – October 2025, 132 months; the 11 erroneous days of 2023 (Lecture 1) excluded
- Limits, before any chart
  - transactions are not users: one firm can send many transactions; exchange users trade off-chain
  - two series with trends can give a high R^2 without a stable link (spurious regression)

Market Value against Metcalfe Value



- Metcalfe value: $\exp(\hat{c} + 2 \ln n_t)$, with b fixed at 2 and $\hat{c}$ set so that the ratio is 1 on average in logs
- The ratio was below 0.5 in 2015–2016 and above 4 in 2021 and 2025 (maximum 9.1, June 2025)
- The ratio drifts upwards: in this sample value grew faster than transactions
- The ratio is a diagnostic that depends on the model ($b = 2$) and on the proxy (transactions), not a fair value

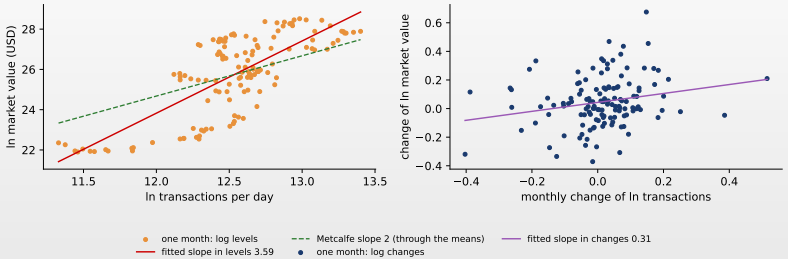
BA_L2_metcalfe ▶ Interactive

Metcalfe's Law: the Estimates

Sample	months	$\hat{b}$	standard error	R^2
2014–2025	132	3.59	0.43	0.58
2014–2019	62	2.71	0.35	0.52
2020–2025	70	1.17	0.48	0.20
2014–2025, with a linear time trend	132	0.27	0.47	0.89
2014–2025, monthly changes $\Delta \ln M_t$ on $\Delta \ln n_t$	131	0.31	0.15	0.05

- Newey–West standard errors (12 lags in levels, 3 in changes)
- Levels, full sample: test of $b = 2$, $t = (3.59 - 2)/0.43 = 3.66$; rejected at 5%
- But the levels regression may be spurious: Engle–Granger test, no cointegration (p-value 0.57)
  - with a time trend or in monthly changes, $\hat{b}$ falls to about 0.3
- Since 2020 the link is weak ($R^2 = 0.20$); ETFs, institutional flows and macro news are plausible drivers, not identified here

Metcalfe's Law in Levels and in Changes



- Left: log levels, 132 months; fitted slope 3.59, against Metcalfe's slope of 2 (dashed green)
- Right: monthly log changes, 131 months; slope 0.31 (SE 0.15), $R^2 = 0.05$, correlation 0.22
- Both levels trend upwards, so their cloud is long and tilted; changes remove the trends, and the link almost disappears
- A relation that holds in levels but not in changes is the warning sign of a spurious regression

BA_L2_metcalfe_changes

Summary

- Annualise with the market's own calendar: $\sigma_a = s\sqrt{365}$ gives 64.1% for Bitcoin, against 19.2% for the S&P 500
- Normality is rejected (Jarque–Bera $> 10\,000$ for all four assets); crash days are far more frequent than the Normal distribution allows
- Volatility clusters: Ljung–Box $Q(10) = 212.7$ for r_t^2, against 20.2 for r_t
- Since 2020 Bitcoin co-moves with equities, like a risk asset: correlation 0.39 with the S&P 500
- Robust variance-ratio tests do not reject a random walk at 2–10 days, in any sub-period; this is not proof of efficiency
- LPPL fits the 2017 bubble with $R^2 \approx 0.98$, but $\hat{t}_c$ moves with the window: a good in-sample fit does not date the crash
- Metcalfe in levels: $\hat{b} = 3.59$, standard error (SE) 0.43; no cointegration, and in monthly changes $\hat{b} = 0.31$
- Lecture 4 models these facts: volatility clustering (GARCH) and heavy tails (VaR 1%, ES 2.5%)

Interpretation Exercises

1. A treasurer reports Bitcoin's annual volatility as 53.3%. What did she do wrong, and what is the right figure?
2. Jarque–Bera $= 29\,620$ for Bitcoin. State H_0, the conclusion, and the consequence for a VaR computed with the Normal distribution.
3. Bitcoin, 2014–2026: Ljung–Box $Q(10) = 20.2$ (p 0.028) for r_t, 212.7 for r_t^2; robust $Z^*(5) = -0.19$ (p 0.853). Is Bitcoin predictable? In what sense? What do the tests not cover?
4. LPPL fit on 1 January – 1 December 2017: $R^2 = 0.98$, $\hat{t}_c =$ 25 December 2017. Would you have sold on 1 December? What does a high R^2 not tell you?
5. Metcalfe: $\hat{b} = 3.59$ (SE 0.43) in levels on 2014–2025, no cointegration (p 0.57), $\hat{b} = 0.31$ (SE 0.15) in monthly changes. What does this imply for valuations based on user growth?

Watch Robust Markowitz for Crypto-Assets

Appendix: Further Material

Optional material for self-study.

- The Tail Index and the Hill Estimator
- Tail Indices of Crypto, Equities and Gold
- Tail Plot and Hill Plot
- Stability of Kurtosis over Time
- Weekday and Weekend Volatility of Bitcoin
- Three Crisis Episodes
- Bitcoin as a Hedge, Safe Haven or Risk Asset
- Average Returns on the Worst Equity Days
- Estimating the LPPL
- LPPL Estimates: Stability of $\hat{t}_c$
- A Rolling Metcalfe Elasticity
- Measuring Volatility Through Time
- Rolling Correlation
- The Robust Variance-Ratio Statistic
- Worked Example: Variance Ratio from Autocorrelations
- The Variance-Ratio Profile
- Bitcoin's Large Drawdowns
- Worked Example: Accelerating Oscillations

The Tail Index and the Hill Estimator

- Far in the tail, losses follow a power law:

$$P(L > x) \approx c\, x^{-\alpha} \quad \text{for large } x$$

  - $L = -r$: the daily loss; $c > 0$: a constant; α: the **tail index**
  - smaller α = heavier tail; moments of order $q \geq \alpha$ are infinite: $\alpha < 4$ means infinite kurtosis
- Hill (1975) estimator from the k largest losses $L_{(1)} \geq L_{(2)} \geq \cdots \geq L_{(k+1)}$, with a confidence interval (CI):

$$\hat{\alpha} = \left[\frac{1}{k} \sum_{i=1}^{k} \ln \frac{L_{(i)}}{L_{(k+1)}} \right]^{-1}, \qquad \mathrm{CI}_{95\%} = \hat{\alpha}\left(1 \pm \frac{1.96}{\sqrt{k}}\right)$$

  - $L_{(k+1)}$: the threshold of the tail; here $k = 2.5\%$ of n
  - large ratios $L_{(i)}/L_{(k+1)}$ on average give a small $\hat{\alpha}$: a heavy tail

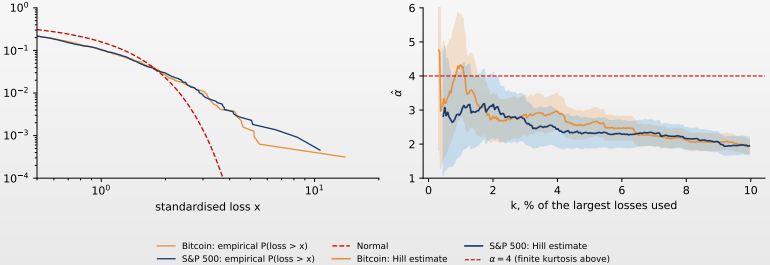
Tail Indices of Crypto, Equities and Gold

	left tail $\hat{\alpha}$	95% CI	right tail $\hat{\alpha}$	95% CI
Bitcoin	2.71	[2.20, 3.21]	3.37	[2.74, 4.00]
Ethereum	2.62	[2.10, 3.15]	3.09	[2.48, 3.71]
S&P 500	2.68	[2.08, 3.29]	2.80	[2.17, 3.43]
Gold	3.11	[2.42, 3.80]	3.04	[2.36, 3.71]

- Daily returns since 2014 (Bitcoin, S&P 500, gold) and 2016 (Ethereum), to 18 September 2026
- Interpretation
  - all left-tail indices are below 4: the sample kurtosis of the four-asset table is not a stable number (next slides)
  - crypto is more *volatile*, but the *shape* of its tails is close to that of equities; the intervals overlap
  - for Bitcoin, losses have a heavier tail than gains (2.71 against 3.37)

SFM_ch14_asset_class

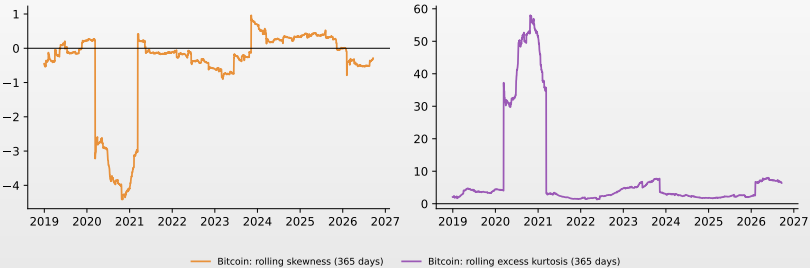
Tail Plot and Hill Plot



- Left: $P(\text{loss} > x)$ on log-log scales; a power-law tail is a straight line with slope $-\alpha$, the Normal (red) bends down
- Right: Hill estimate against k with 95% bands, 2018–2026. Bitcoin: 4.33 ($k = 1\%$), 2.71 (2.5%), 2.60 (5%), 1.94 (10%); S&P 500: 2.92, 2.81, 2.32, 1.95
- Small k: few observations, wide bands; large k: the centre of the distribution biases $\hat{\alpha}$ down. Report $\hat{\alpha}$ on the flat part, here $k \approx 2.5\%$

BA_L2_tails

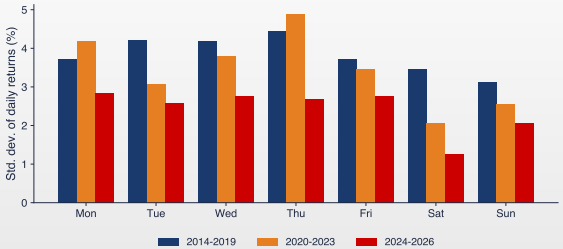
Stability of Kurtosis over Time



- Skewness and excess kurtosis of Bitcoin on rolling 365-day windows
- Kurtosis ranges from 1.3 to 58.0 (window ending October 2020, which contains 12 March 2020); today 6.4
- One day moves the estimate by a factor of ten: with $\alpha < 4$ the sample kurtosis has no stable limit (previous slides)

BA_L2_rolling_moments

Weekday and Weekend Volatility of Bitcoin



- Standard deviation of Bitcoin's daily returns by closing day, three periods
- No weekend effect in the *mean*; but weekend variance is only 35–66% of weekday variance
- Crypto trades on Saturdays, but banks, macro news and institutional traders do not: risk follows the traditional calendar

SFM_ch14_asset_class

Three Crisis Episodes

Episode	period	Bitcoin	Ethereum	S&P 500	gold
COVID-19	19 Feb – 23 Mar 2020	−40.6	−65.5	−41.4	0.3
Terra/Luna	5 – 12 May 2022	−23.0	−33.8	−5.4	−3.0
FTX	5 – 21 Nov 2022	−29.9	−38.4	4.6	3.4

- Cumulative log returns over each episode, in log-return percentage points
  - as simple returns $e^{r/100} - 1$: COVID-19 Bitcoin −33.4%, Ethereum −48.1%, S&P 500 −33.9%
- Interpretation
  - a global shock (COVID-19) hits crypto as hard as equities; gold does not move
  - crypto-specific shocks (the collapse of the stablecoin TerraUSD; the bankruptcy of the exchange FTX) cost Bitcoin and Ethereum at least 20% (simple returns), equities hardly anything
  - in these three episodes the link is **asymmetric**: the equity crash reached crypto, the crypto crashes did not reach equities

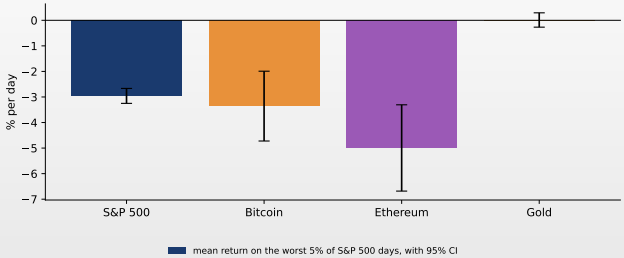
Bitcoin as a Hedge, Safe Haven or Risk Asset

- Baur & Lucey (2010): a **hedge** is uncorrelated or negatively correlated with equities on average; a **safe haven** is uncorrelated or negatively correlated with equities in extreme market stress
- A simpler check here: the mean return of asset x on the set $\mathcal{S}$ of the worst 5% of S&P 500 days

$$\bar{x}_{\mathcal{S}} = \frac{1}{n_{\mathcal{S}}} \sum_{t \in \mathcal{S}} x_t, \qquad t = \frac{\bar{x}_{\mathcal{S}}}{s_{\mathcal{S}} / \sqrt{n_{\mathcal{S}}}}$$

  - 2018–2026, common days: $n_{\mathcal{S}} = 110$ days with the S&P 500 below -1.77%, on average -2.96%
- Means with 95% CI: Bitcoin -3.36% $[-4.72, -1.99]$, Ethereum -4.99% $[-6.68, -3.31]$, gold $+0.01\%$ $[-0.27, 0.29]$
- Reading
  - on these days crypto falls on average more than equities
  - gold: a zero mean is not rejected; this does not show that its mean is non-negative, nor that gold is a safe haven in the sense of Baur & Lucey (2010)

Average Returns on the Worst Equity Days



- Mean daily return on the 110 worst S&P 500 days of 2018–2026, with 95% confidence intervals
- For a corporate treasury: in this sample, Bitcoin lost on average on the days when equities were under stress

BA_L2_safe_haven

Estimating the LPPL

- Rewrite the cosine term with $C_1 = C\cos\phi$ and $C_2 = C\sin\phi$:

$$\ln P_t = A + Bf_t + C_1 g_t + C_2 h_t + \varepsilon_t$$

$$f_t = (t_c - t)^m, \quad g_t = (t_c - t)^m \cos(\omega \ln(t_c - t)), \quad h_t = (t_c - t)^m \sin(\omega \ln(t_c - t))$$

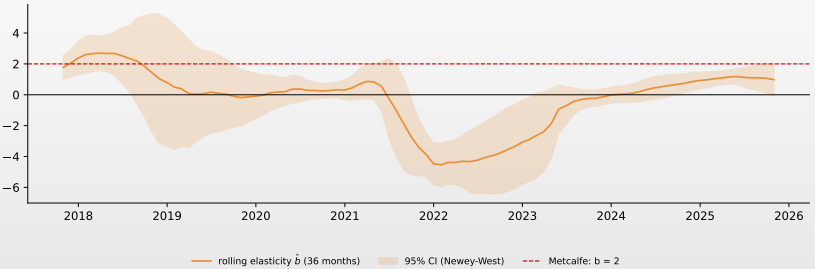
- Two steps
  1. for given (t_c, m, ω), the model is linear in (A, B, C_1, C_2): ordinary least squares
  2. search over (t_c, m, ω) to minimise the sum of squared errors $\sum_t \varepsilon_t^2$
- Caution
  - many local minima: start the search from many points; fit on many windows
  - t_c is the least precise parameter; fits look convincing after the crash, real-time warnings give false alarms

LPPL Estimates: Stability of $\hat{t}_c$

window ends	n	$\hat{t}_c$	lead (days)	$\hat{m}$	$\hat{\omega}$	$\lvert\hat{C}\rvert/\lvert\hat{B}\rvert$	R^2
15 Nov 2017	319	4 Jan 2018	50	0.900	13.00	0.089	0.980
1 Dec 2017	335	25 Dec 2017	24	0.851	12.49	0.088	0.982
8 Dec 2017	342	8 Dec 2017	0	0.741	11.14	0.081	0.979

- All windows start on 1 January 2017; $\hat{B} \approx -2.8$ in all three: super-exponential growth
- Interpretation
  - $R^2 \approx 0.98$ in every window: a good fit does not mean a reliable forecast
  - $\hat{t}_c$ moves by four weeks when the window moves by three weeks
  - 15 November: $\hat{m}$ and $\hat{\omega}$ sit on the bounds of the search (0.9 and 13), a sign of a poorly identified fit
  - 8 December: $\hat{t}_c$ equals the last observation, a degenerate solution
- Practice: fit many windows and report the share of windows that pass the filters (the log-periodic power law singularity (LPPLS) confidence indicator)

A Rolling Metcalfe Elasticity



- $\hat{b}$ on rolling 36-month windows with Newey–West 95% intervals; red: Metcalfe's $b = 2$
- $\hat{b}$ peaks at 2.70 (window ending March 2018), turns negative in 2021–2023 (value fell while transactions grew), ends at 0.98 (October 2025)
- The interval covers 2 in only 28% of the windows: there is no stable network-value elasticity

BA_L2_metcalfe_rolling

Measuring Volatility Through Time

- **Rolling volatility** on a window of the last w days, annualised:

$$\hat{\sigma}_{a,t} = \sqrt{A}\sqrt{\frac{1}{w-1}\sum_{j=0}^{w-1}\left(r_{t-j} - \bar{r}_t\right)^2}, \qquad \bar{r}_t = \frac{1}{w}\sum_{j=0}^{w-1} r_{t-j}$$

  - w: the window length (30 days for crypto, 21 trading days for equities, about one month)
  - A: returns per year (365 or 252); the window moves forward one day at a time
- Trade-off in the choice of w
  - short window: reacts fast, but noisy; long window: smooth, but late
  - the standard error of $\hat{\sigma}$ is roughly $\hat{\sigma}/\sqrt{2w}$: about 13% of $\hat{\sigma}$ for $w = 30$
- Lecture 4 replaces the window by a model: exponentially weighted moving average (EWMA) or GARCH

Rolling Correlation

- Pearson correlation of two return series x (Bitcoin) and y (S&P 500) on the last w common days:

$$\hat{\rho}_{xy,t} = \frac{\sum_{j=0}^{w-1}(x_{t-j} - \bar{x}_t)(y_{t-j} - \bar{y}_t)}{\sqrt{\sum_{j=0}^{w-1}(x_{t-j} - \bar{x}_t)^2}\sqrt{\sum_{j=0}^{w-1}(y_{t-j} - \bar{y}_t)^2}}$$

  - $w = 250$ common days, about one year of equity trading
  - standard error $\approx (1 - \rho^2)/\sqrt{w} \approx 0.06$: changes below 0.1 are noise
- Two calendars
  - join *prices* on common days first, then compute returns: Monday's Bitcoin return then covers Friday to Monday, like the equity return
  - closes are not simultaneous (midnight UTC, Coordinated Universal Time, vs 16:00 New York): weekly returns give a check

The Robust Variance-Ratio Statistic

- Volatility clustering makes the independent and identically distributed (i.i.d.) test too eager to reject; Lo & MacKinlay (1988) give a robust version:

$$Z^*(q) = \frac{\widehat{\mathrm{VR}}(q) - 1}{\sqrt{\hat{\theta}^*(q)}} \;\approx\; N(0,1)$$

$$\hat{\theta}^*(q) = \sum_{j=1}^{q-1} \left[\frac{2(q-j)}{q}\right]^2 \hat{\delta}_j, \qquad \hat{\delta}_j = \frac{\sum_{t=j+1}^{n} (r_t - \bar{r})^2 (r_{t-j} - \bar{r})^2}{\left[\sum_{t=1}^{n} (r_t - \bar{r})^2\right]^2}$$

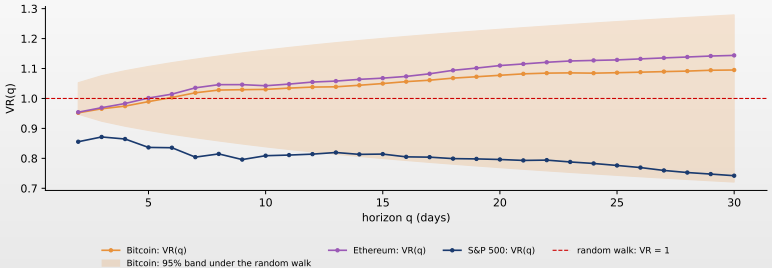
  - $\hat{\delta}_j$: the variance of the lag-j autocorrelation when volatility changes over time
  - reject the random walk at 5% if $|Z^*(q)| > 1.96$

- Link with autocorrelations: $\mathrm{VR}(q) \approx 1 + 2\sum_{j=1}^{q-1} \left(1 - \frac{j}{q}\right)\rho(j)$
  - positive autocorrelations push VR above 1, negative ones below 1

Worked Example: Variance Ratio from Autocorrelations

- For $q = 2$: $\mathrm{VR}(2) \approx 1 + 2\,(1 - \tfrac{1}{2})\,\rho(1) = 1 + \rho(1)$
- Bitcoin, since 2014: $\hat{\rho}(1) = -0.022$, so $\mathrm{VR}(2) \approx 0.978$; the direct estimate is 0.977
- S&P 500, same period: $\hat{\rho}(1) = -0.126$, so $\mathrm{VR}(2) \approx 0.874$; the direct estimate is 0.875
- Interpretation
  - the S&P 500 index shows mild daily mean reversion, typical of an index (bid-ask bounce, stale prices of its members)
  - the robust test does not reject because the variance of $\widehat{\mathrm{VR}}$ is large when volatility clusters

The Variance-Ratio Profile



- $\widehat{\mathrm{VR}}(q)$ for $q = 2, \ldots, 30$ days, 2018–2026; shaded: the robust 95% band for Bitcoin
- Bitcoin: VR(5) = 0.989 ($Z^* = -0.19$), VR(20) = 1.077 ($Z^* = 0.66$); Ethereum VR(20) = 1.11 ($Z^* = 0.95$)
- S&P 500: VR(20) = 0.796 ($Z^* = -0.76$): mean reversion in the point estimate, but not significant

BA_L2_vr_profile

Bitcoin's Large Drawdowns

Peak	USD	trough	USD	depth (%)	new record	days under water
2014-09-17	457	2015-01-14	178	−61.1	2015-12-15	454
2017-12-16	19 497	2018-12-15	3 237	−83.4	2020-11-30	1 080
2021-04-13	63 503	2021-07-20	29 807	−53.1	2021-10-19	189
2021-11-08	67 567	2022-11-21	15 787	−76.6	2024-03-04	847
2025-10-06	124 753	2026-06-30	58 559	−53.1	not yet	–

- Episodes since September 2014 deeper than 50%
- Interpretation
  - five falls of more than half in twelve years: a crash of this size is a regular event
  - the 2017 and 2021 peaks took 1 080 and 847 days to recover

SFM_ch14_bubbles_etf

Worked Example: Accelerating Oscillations

- Peaks of the cosine term satisfy $\omega \ln(t_c - t_k) - \phi = 2\pi k$, so consecutive distances to t_c shrink by a constant factor

$$\frac{t_c - t_{k+1}}{t_c - t_k} = e^{-2\pi/\omega}$$

- Bitcoin 2017 fit (slide "Bitcoin 2017: an LPPL Fit"): $\hat{\omega} = 12.49$, so $e^{-2\pi/12.49} = 0.605$
  - a peak 30 days before t_c is followed by peaks 18.1, 11.0 and 6.6 days before t_c
- The ratio $\lambda = e^{2\pi/\omega} = 1.65$ is the "preferred scaling ratio" of the hierarchy of traders in Johansen, Ledoit & Sornette (2000)
- Practical reading: shrinking gaps between local tops are the visual signature of an LPPL bubble